# Frontiers of Theoretical Physics

Lecturers:
David Mateos (coordinator) and Joan Solà

# Frontiers of theoretical physics

- Renormalization group.
- Introduction to supersymmetry.
- Gauge/string correspondence.

**Part 1:**
**David Mateos**

- Grand unified theories.
- Phenomenology of supersymmetric theories.
- Open problems in cosmology.

**Part II:**
**Joan Solà**

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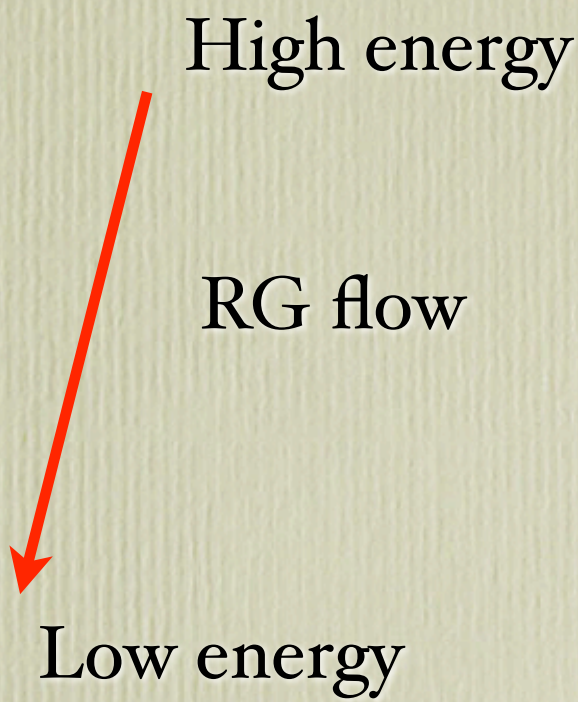
**Recommended background:**

- Basic GR and QFT.
- Elementary Particles
- Standard Model (simultaneous)

# Renormalization group

David Mateos

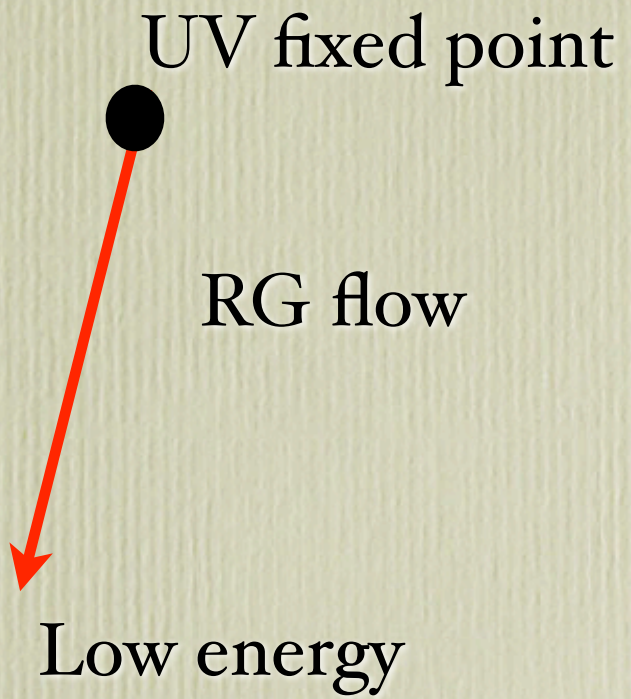
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# Renormalization group

David Mateos

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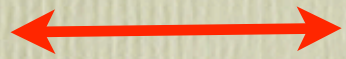



- Modern definition of QFT.

# Introduction to supersymmetry

David Mateos

- Fundamental symmetry:

Bosons  Fermions

- Essential ingredient in e.g. string theory.

# Phenomenology of supersymmetric theories

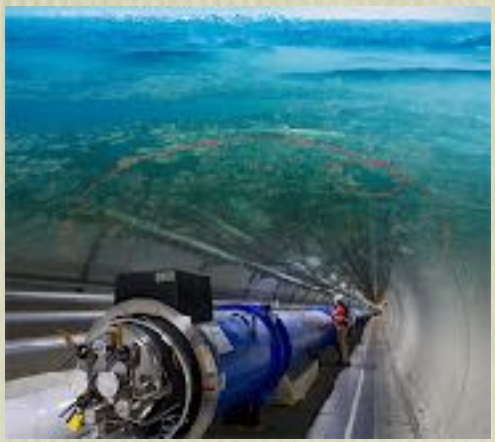
Joan Solà

- Supersymmetry is a nice theory, but how do we “see” it?

# Phenomenology of supersymmetric theories

Joan Solà

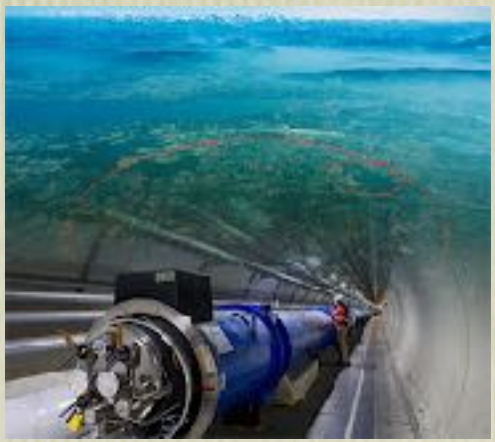
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- Supersymmetry predicts new particles!
  - Direct SUSY signals: Find new particles.



# Phenomenology of supersymmetric theories

Joan Solà

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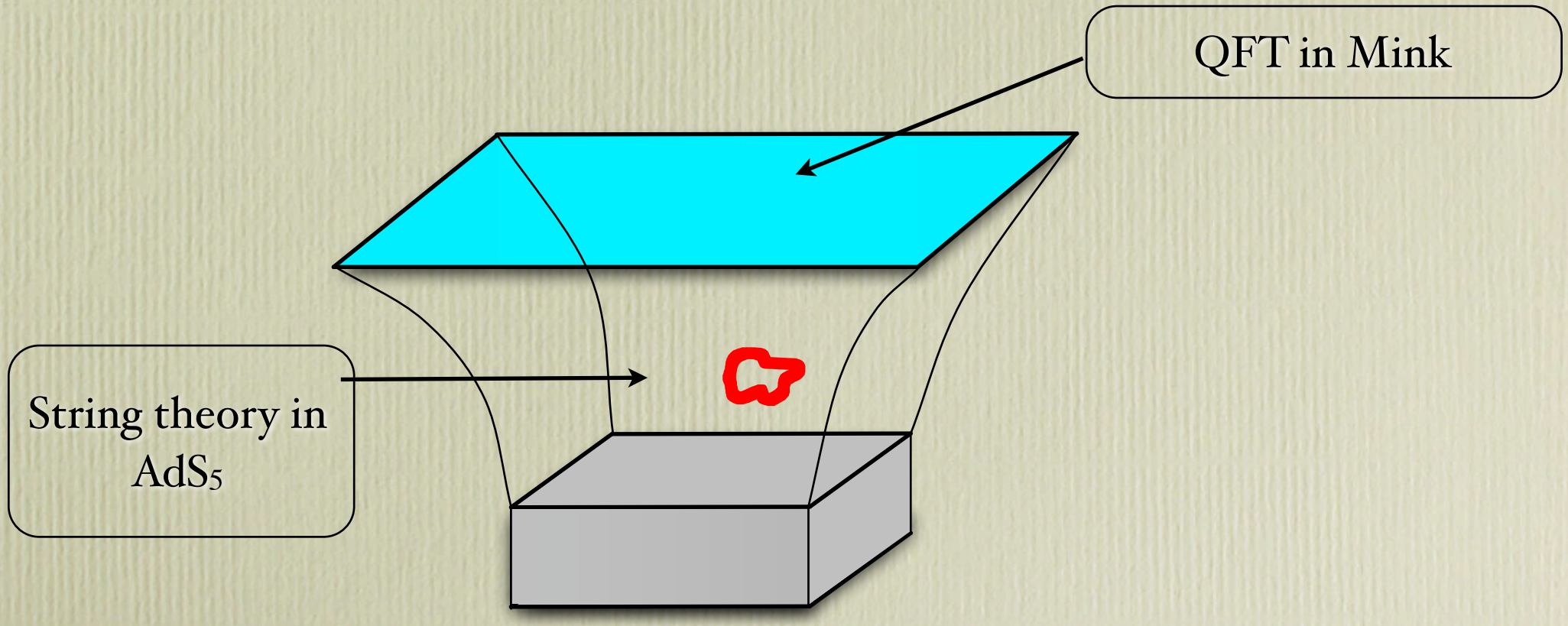


  - Direct SUSY signals: Find new particles.
  - Indirect SUSY signals: quantum effects to precision physics.

# Gauge/gravity correspondence

David Mateos

- Profound equivalence.
- Best definition of Quantum Gravity:



# Grand unified theories



- Standard Model has 3 independent interactions (couplings).
- Is it possible to unify some (or all) the couplings?

# Grand unified theories

Joan Solà

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- Example:
  - Electricity: $\dfrac{1}{4\pi\epsilon_0}$
  - Magnetism: $\dfrac{\mu_0}{4\pi}$
  - Maxwell's Electromagnetism: $\mu_0\epsilon_0 = \dfrac{1}{c^2}$

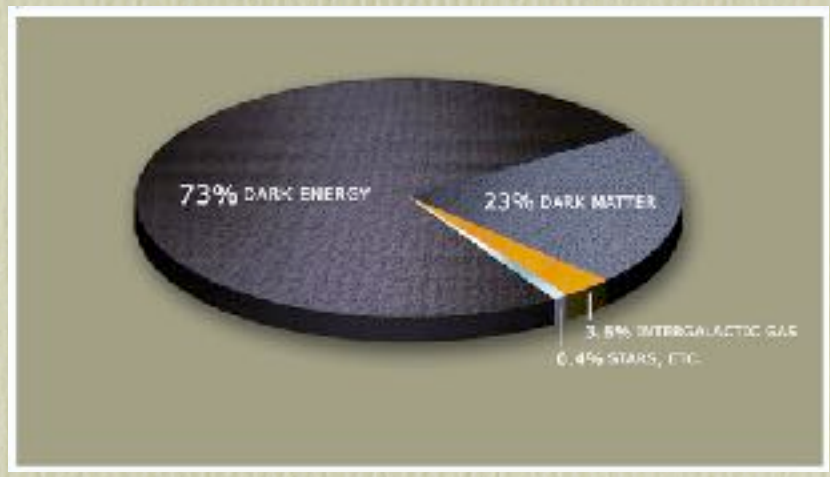
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- Unification conditions for SM group.
- New physical phenomena: proton decay.
- Unification conditions in SUSY models.

# Open problems in cosmology

Joan Solà




- Cosmological constant and dark energy: beyond the standard LCDM model.
- Current weaknesses and tensions of the standard LCDM model.
- Dynamical vacuum energy models.
- Quintessence and phantom dynamical dark energy.
- Dark matter and WIMPS: candidates and thermal history in the early universe.
- Vacuum energy in QFT, formal aspects.

# Practical details

- **Language**: English.
- **Schedule**: Mon-Thu from 11:40 to 12:40, room V12M.
- **Duration (approximately):** 12 weeks from mid-Feb to mid-May.
- **Distribution:** 6 weeks for each part
- **Evaluation**:
  - Exercise sheets and/or final interview and/or final exam.
    Sheets must be handed in on time or they will NOT be accepted!
  - Must pass each part separately.
  - If so then the final grade is the average.
- **Re-evaluation**: In September with similar rules.
- **Bibliography and detailed syllabus:** Each lecturer will tell you.

Questions welcome!